# Improved zero-density estimates for the Riemann zeta function

Mark Lewko

We establish improved zero-density estimates for the Riemann zeta function relative to the bounds previously released in PALOMAR-2026-10-11-000010 v1 and PALOMAR-2026-10-09-000005 v1. Let N(σ,T) count the nontrivial zeros with Re ρ ≥ σ and |Im ρ| ≤ T, with analytic multiplicity. For T ≥ 2 and ε > 0, our bounds have the form $N(\sigma,T) \le C(\sigma,\varepsilon)T^{(1-\sigma)A(\sigma)+\varepsilon}$. The native quantitative bounds improve those earlier power curves for 12/17 < σ ≤ 3/4 and 7/9 < σ ≤ 7/8. The displayed power-curve comparison continues for 7/8 < σ < 17/19, where the imported OpenAI result already gives N(σ,T)=0. Later Lean-formalized refinements also give a zero-free region below 7/8 [Collet26]; our comparisons near 7/8 concern the named earlier results. Table 1 gives all the proved contribution curves and their full ranges, including A(σ)=12/(9σ−1) on [5/7,113/153] and A(σ)=255/(153σ+7) on [113/153,3/4]. Figure 1 compares the cumulative curves.

$$N(\sigma, T) \le C_{\sigma,\varepsilon}\, T^{p(\sigma)+\varepsilon}, \qquad p(\sigma) = A(\sigma)(1-\sigma)$$

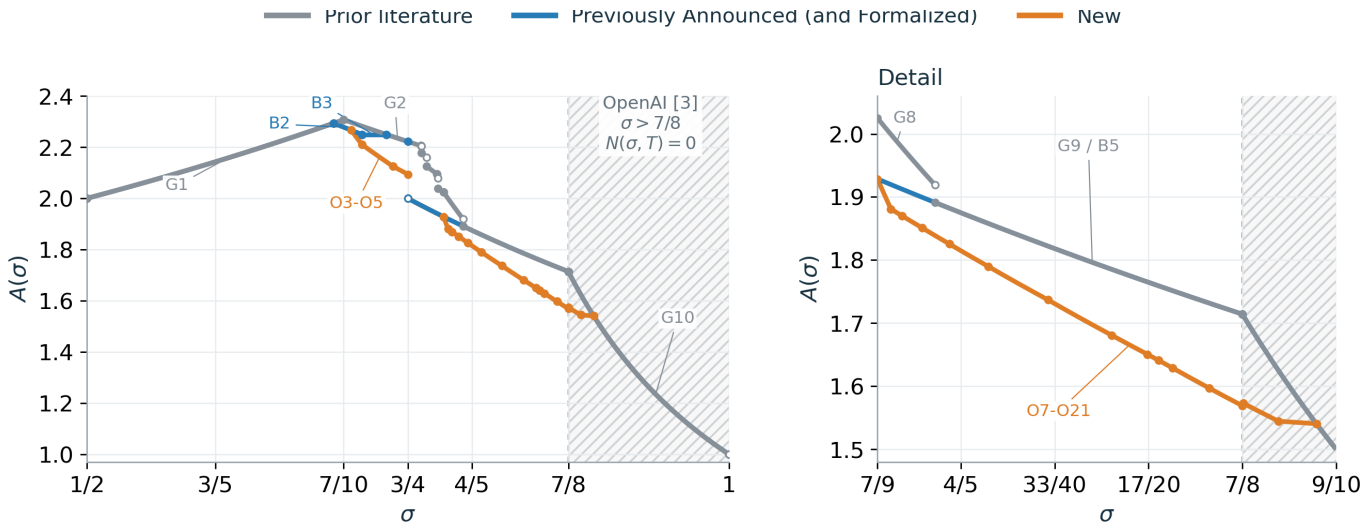



Figure 1. The three curves are cumulative minima of the listed bounds: Prior literature (gray), Previously Announced (and Formalized) (blue), and New (orange). Coincident portions use the earlier color. Exact formulas and endpoints are listed on pages 4-5.

Neutral hatching cites OpenAI [3]: N(σ,T)=0 for σ > 7/8; the boundary is excluded. This checked input is credited to OpenAI. Segment IDs and exact formulas are on pages 4-5.

The plotted power curves compare the named earlier submissions; “New” denotes additions to that comparison. Hatching marks the imported OpenAI input at σ>7/8. Later zero-free refinements are cited in the text.

The 73 main density statements passed protected Comparator with Lean, NanoDa and verified con-ron. Page 6 reports separate 4.34.1 supplements. References: [1], [2], [3].

# Table 1. Proved contribution bounds

Write p(σ) = (1 - σ)A(σ), except where p is given directly. S means C may depend on fixed σ and ε. U means one Cε works for every σ in the printed range. In each case C is chosen before all real T ≥ 2.

| ID | A(σ), or p(σ) | Full σ range | C | Attribution |
|---|---|---|---|---|
| P11 | $A = \frac{18}{7\sigma+3}$ | $\left[\frac{9}{13}, \frac{5}{7}\right]$ | S | Lewko<br>announced |
| P12 | $A = \frac{9}{4}$ | $\left[\frac{5}{7}, \frac{3}{4}\right]$ | S | Lewko<br>announced |
| P13 | $A = \frac{3}{2\sigma}$ | $\left(\frac{3}{4}, 1\right)$ | S | Lewko<br>announced |
| P14 | $A = \frac{6}{8\sigma-3}$ | $\left[\frac{7}{10}, \frac{5}{7}\right]$ | S | Lewko †<br>domination / native |
| P15 | $A = \frac{12}{9\sigma-1}$ | $\left[\frac{5}{7}, \frac{113}{153}\right]$ | S | Lewko<br>new |
| P16 | $A = \frac{255}{153\sigma+7}$ | $\left[\frac{113}{153}, \frac{3}{4}\right]$ | S | Lewko<br>new |
| P17 | $A = \frac{3}{11\sigma-7}$ | $\left[\frac{7}{9}, \frac{29359}{37574}\right]$ | S | Lewko<br>new |
| P18 | $A = \frac{4744}{4870\sigma-1283}$ | $\left[\frac{29359}{37574}, \frac{1015}{1294}\right]$ | S | Lewko<br>new |
| P19 | $A = \frac{6154}{6354\sigma-1693}$ | $\left[\frac{1015}{1294}, \frac{353}{447}\right]$ | S | Lewko<br>new |
| P20 | $A = \frac{3782}{3936\sigma-1065}$ | $\left[\frac{353}{447}, \frac{467}{586}\right]$ | S | Lewko<br>new |
| P21 | $A = \frac{2168}{2282\sigma-631}$ | $\left[\frac{467}{586}, \frac{289}{358}\right]$ | S | Lewko<br>new |
| P22 | $A = \frac{566}{606\sigma-173}$ | $\left[\frac{289}{358}, \frac{163}{198}\right]$ | S | Lewko<br>new |
| P23 | $A = \frac{31544}{35670\sigma-11203}$ | $\left[\frac{163}{198}, \frac{7873}{9370}\right]$ | S | Lewko<br>new |
| P24 | $A = \frac{1261710}{1447460\sigma-465509}$ | $\left[\frac{7873}{9370}, \frac{2018}{2375}\right]$ | S | Lewko<br>new |
| P25 | $A = \frac{3066}{3616\sigma-1215}$ | $\left[\frac{2018}{2375}, \frac{1959705}{2298302}\right]$ | S | Lewko<br>new |
| P26 | $A = \frac{597962}{729216\sigma-257415}$ | $\left[\frac{1959705}{2298302}, \frac{6482825}{7570038}\right]$ | S | Lewko<br>new |
| P27 | $A = \frac{16504356}{20732766\sigma-7623595}$ | $\left[\frac{6482825}{7570038}, \frac{2905}{3354}\right]$ | S | Lewko<br>new |
| P28 | $A = \frac{2172}{2742\sigma-1015}$ | $\left[\frac{2905}{3354}, \frac{7}{8}\right]$ | S | Lewko<br>new |
| P29 | $p = \frac{1086}{5537}$ | $\left[\frac{7}{8}, 1\right]$ | U | Lewko<br>monotonicity |
| P30 | $A = \frac{24}{30\sigma-11}$ | $\left[\frac{19}{22}, \frac{23}{26}\right]$ | U | Lewko †<br>native / zero-free |
| P31 | $A = \frac{6}{\sigma+3}$ | $\left[\frac{23}{26}, 1\right]$ | U | Lewko †<br>zero-free consequence |

**Also proved: the previously announced global bound**

A(σ) = 39/17 on the closed interval [1/2,1], with a uniform Cε. This weaker bound is retained here and is inactive in the cumulative frontier.

† Existing-bound or zero-free consequences; see page 3. All printed bounds and ranges remain valid.

# Historical comparisons and constants

These ten rows are comparisons with prior literature. The full proved source ranges are shown, even where the corresponding curve is not the minimum.

| ID | A(σ), or p(σ) | Full σ range | C | Attribution |
|---|---|---|---|---|
| P1 | $A = \frac{3}{2-\sigma}$ | $\left[\frac{1}{2}, \frac{7}{10}\right]$ | S | Ingham |
| P2 | $A = \frac{15}{5\sigma+3}$ | $\left[\frac{7}{10}, \frac{19}{25}\right)$ | S | Guth-Maynard |
| P3 | $A = \frac{9}{8\sigma-2}$ | $\left[\frac{19}{25}, \frac{127}{167}\right)$ | S | Ivić / TTY Table 2 |
| P4 | $A = \frac{15}{13\sigma-3}$ | $\left[\frac{127}{167}, \frac{13}{17}\right)$ | S | TTY Table 2 row 4 |
| P5 | $A = \frac{6}{5\sigma-1}$ | $\left[\frac{13}{17}, \frac{17}{22}\right)$ | S | TTY Table 2 row 5 |
| P6 | $A = \frac{2}{9\sigma-6}$ | $\left[\frac{17}{22}, \frac{41}{53}\right)$ | S | TTY Table 2 row 6 |
| P7 | $A = \frac{9}{7\sigma-1}$ | $\left[\frac{41}{53}, \frac{7}{9}\right)$ | S | TTY Table 2 row 7 |
| P8 | $A = \frac{9}{16\sigma-8}$ | $\left[\frac{7}{9}, \frac{1867}{2347}\right)$ | S | TTY Table 2 row 8 |
| P9 | $A = \frac{3}{2\sigma}$ | $\left[\frac{23}{29}, 1\right]$ | S | Kerr Thm. 6 |
| P10 | $A = \frac{3}{10\sigma-7}$ | $\left(\frac{7}{10}, 1\right]$ | S | Heath-Brown |

## Constant dependence

S: for each fixed σ in the range and each ε > 0, there is C(σ,ε) > 0 such that the bound holds for every real T ≥ 2.

U: for each ε > 0, there is Cε > 0 such that the bound holds simultaneously for every σ in the printed range and every real T ≥ 2. This is proved for P29-P31 and the global 39/17 bound.

The announced 9/4 row is stated with C(σ,ε); no uniform constant is asserted for that row.

Square brackets include the endpoint; round brackets exclude it. At σ = 1 the actual zero count is zero. The plotted normalized A is a left limit.

## Attribution and proof routes

Historical comparisons use stronger native bounds and OpenAI [3] where applicable. The curve A=6/(8σ-3) uses announced-bound domination on [7/10,12/17] and the native two-term bound on (12/17,5/7]. The curve A=24/(30σ-11) uses native domination through 7/8 and zero-freeness above it. A=6/(σ+3) on [23/26,1] is a zero-free consequence. Historical rows A=3/(2σ) and A=3/(10σ-7) also use zero-freeness above 7/8. These are checked conclusions; the original historical arguments are not claimed here.

# Exact frontier: prior and announced

The IDs agree with the figure. These are active intervals, not the full source ranges on pages 2-3. An endpoint may be supplied by an equal-valued adjacent theorem. At σ = 1 only the left limit of A is plotted.

| IDs | A(σ) | Active σ interval |
|---|---|---|
| G1<br>P1 | $\frac{3}{2-\sigma}$ | $[\frac{1}{2}, \frac{7}{10}]$ |
| G2<br>P2 | $\frac{15}{5\sigma+3}$ | $[\frac{7}{10}, \frac{19}{25}]$ |
| G3<br>P3 | $\frac{9}{8\sigma-2}$ | $[\frac{19}{25}, \frac{127}{167})$ |
| G4<br>P4 | $\frac{15}{13\sigma-3}$ | $[\frac{127}{167}, \frac{13}{17})$ |
| G5<br>P5 | $\frac{6}{5\sigma-1}$ | $[\frac{13}{17}, \frac{17}{22}]$ |
| G6<br>P6 | $\frac{2}{9\sigma-6}$ | $[\frac{17}{22}, \frac{41}{53})$ |
| G7<br>P7 | $\frac{9}{7\sigma-1}$ | $[\frac{41}{53}, \frac{7}{9}]$ |
| G8<br>P8 | $\frac{9}{16\sigma-8}$ | $[\frac{7}{9}, \frac{23}{29})$ |
| G9<br>P9 | $\frac{3}{2\sigma}$ | $[\frac{23}{29}, \frac{7}{8}]$ |
| G10<br>P10 | $\frac{3}{10\sigma-7}$ | $[\frac{7}{8}, 1)$ |
| B1<br>P1 | $\frac{3}{2-\sigma}$ | $[\frac{1}{2}, \frac{9}{13}]$ |
| B2<br>P11 | $\frac{18}{7\sigma+3}$ | $[\frac{9}{13}, \frac{5}{7}]$ |
| B3<br>P12 | $\frac{9}{4}$ | $[\frac{5}{7}, \frac{11}{15}]$ |
| B4<br>P2 | $\frac{15}{5\sigma+3}$ | $[\frac{11}{15}, \frac{3}{4}]$ |
| B5<br>P13 | $\frac{3}{2\sigma}$ | $(\frac{3}{4}, \frac{7}{8}]$ |
| B6<br>P10 | $\frac{3}{10\sigma-7}$ | $[\frac{7}{8}, 1)$ |

# Exact frontier: new cumulative bounds

The IDs agree with the figure. These are active intervals, not the full source ranges on pages 2-3. An endpoint may be supplied by an equal-valued adjacent theorem. At σ = 1 only the left limit of A is plotted.

| IDs | A(σ) | Active σ interval |
|---|---|---|
| O1<br>P1 | $\frac{3}{2-\sigma}$ | $[\frac{1}{2}, \frac{9}{13}]$ |
| O2<br>P11 | $\frac{18}{7\sigma+3}$ | $[\frac{9}{13}, \frac{12}{17}]$ |
| O3<br>P14 | $\frac{6}{8\sigma-3}$ | $[\frac{12}{17}, \frac{5}{7}]$ |
| O4<br>P15 | $\frac{12}{9\sigma-1}$ | $[\frac{5}{7}, \frac{113}{153}]$ |
| O5<br>P16 | $\frac{255}{153\sigma+7}$ | $[\frac{113}{153}, \frac{3}{4}]$ |
| O6<br>P13 | $\frac{3}{2\sigma}$ | $(\frac{3}{4}, \frac{7}{9}]$ |
| O7<br>P17 | $\frac{3}{11\sigma-7}$ | $[\frac{7}{9}, \frac{29359}{37574}]$ |
| O8<br>P18 | $\frac{4744}{4870\sigma-1283}$ | $[\frac{29359}{37574}, \frac{1015}{1294}]$ |
| O9<br>P19 | $\frac{6154}{6354\sigma-1693}$ | $[\frac{1015}{1294}, \frac{353}{447}]$ |
| O10<br>P20 | $\frac{3782}{3936\sigma-1065}$ | $[\frac{353}{447}, \frac{467}{586}]$ |
| O11<br>P21 | $\frac{2168}{2282\sigma-631}$ | $[\frac{467}{586}, \frac{289}{358}]$ |
| O12<br>P22 | $\frac{566}{606\sigma-173}$ | $[\frac{289}{358}, \frac{163}{198}]$ |
| O13<br>P23 | $\frac{31544}{35670\sigma-11203}$ | $[\frac{163}{198}, \frac{7873}{9370}]$ |
| O14<br>P24 | $\frac{1261710}{1447460\sigma-465509}$ | $[\frac{7873}{9370}, \frac{2018}{2375}]$ |
| O15<br>P25 | $\frac{3066}{3616\sigma-1215}$ | $[\frac{2018}{2375}, \frac{1959705}{2298302}]$ |
| O16<br>P26 | $\frac{597962}{729216\sigma-257415}$ | $[\frac{1959705}{2298302}, \frac{6482825}{7570038}]$ |
| O17<br>P27 | $\frac{16504356}{20732766\sigma-7623595}$ | $[\frac{6482825}{7570038}, \frac{2905}{3354}]$ |
| O18<br>P28 | $\frac{2172}{2742\sigma-1015}$ | $[\frac{2905}{3354}, \frac{7}{8}]$ |
| O19<br>P29 | $\frac{1086}{5537(1-\sigma)}$ | $[\frac{7}{8}, \frac{24139}{27578}]$ |
| O20<br>P30 | $\frac{24}{30\sigma-11}$ | $[\frac{24139}{27578}, \frac{23}{26}]$ |
| O21<br>P31 | $\frac{6}{\sigma+3}$ | $[\frac{23}{26}, \frac{17}{19}]$ |
| O22<br>P10 | $\frac{3}{10\sigma-7}$ | $[\frac{17}{19}, 1)$ |

# Bounds for separate objects

These are separate Lean 4.34.1 supplements, outside the main 73-statement 4.35 submission. Their source and existing kernel receipts were rechecked, with fresh proof-closure inspection. Both zero subsets lie in the positive closed slab T ≤ Im ρ ≤ 2T. They are not full-density estimates for N(σ,T).

## Polynomial large values at σ = 17/20

$$|W| \le C_\varepsilon T^{\frac{2817}{11744}+\varepsilon}$$

$$\sigma = \frac{17}{20}, \quad T^{\frac{1565}{2936}} \le N \le T^{\frac{4695}{5872}}$$

The polynomial length N lies in the displayed closed range. Coefficients are bounded by 1; translates are arbitrary and the sample points are one-separated, with polynomial modulus at least N to the power 17/20. For each ε > 0, Cε is chosen before all physical data and real T ≥ 2.

## Positive canonical LONG zeros

$$R_{\mathrm{LONG}} \le C\, T^{\frac{395(1-\sigma)}{132(11\sigma-7)}+\varepsilon}$$

$$[L,U] \subset \left(\frac{2243}{2904}, \frac{3301}{4228}\right)$$

For every closed [L,U] strictly inside (2243/2904,3301/4228), choose δ(U) > 0 before η, ε, σ and T. For fixed L and U, C(L,U,η,ε) is then chosen before every σ in [L,U] and every real T ≥ 2. Analytic multiplicity is retained. In particular [7/9,1561/2000] is a closed admissible interval.

## Positive canonical SHORT zeros at σ = 17/20

$$R_{\mathrm{SHORT}} \le C_{\delta,\varepsilon} T^{\frac{2817}{11744}+\varepsilon}$$

$$\sigma = \frac{17}{20}, \quad a = \frac{1565}{2936}$$

The detector exponent is a = 1565/2936. Choose η(ε) = min(1/16384, ε/2404) before each fixed positive smoothing δ; then choose C(δ,ε) before all real T ≥ 2. SHORT means membership in the original admissible normalized powered-polynomial family; LONG is its exact complement. Analytic multiplicity is retained.

# Appendix: source and verification

## Source scope

The 21 contribution rows, ten historical rows, weaker global bound, and three separate-object estimates are driven by the existing checked inventory. No proof source, mathematical range or exponent is changed by this presentation.

## Imported input

OpenAI [3] supplies the strict zero-free half-plane Re s > 7/8. Its imported proof is kernel-checked here and credited to OpenAI. Upstream notices are retained.

## Later literature

Subsequent Lean-verified refinements lower the zero-free boundary below 7/8, including Hailey Collet's strict bound Re(s)>874861/1000000=0.874861 [Collet26]. Liu's intermediate refinement gives B∞=0.874957019420098946... [Liu26]. Our comparison near 7/8 is with the specified earlier results and does not improve the current zero-free region.

## Literature verification scope

The imported, kernel-checked input used by this formalization remains OpenAI's strict 7/8 theorem. Collet's result has separate public tracker verification with Lean, NanoDa and con-ron. Liu's repository reports a strict Lean source replay using pinned external artifacts. These later results are cited as literature; they are not imported into this release's 73-statement proof closure, and no new replay of them was run for this editorial revision.

## Current checks

The 73 main statements on Lean 4.35.0-rc2 / Mathlib 065356127b1dc0016f66b7283ce0ce2c4055aa55 passed protected Comparator on 20 October 2026, including Lean, NanoDa and verified con-ron. Only Classical.choice, propext and Quot.sound are permitted. The proof sources and nine dependency pins are unchanged. Page 6 has separate 4.34.1 supplement coverage.

## Verification scope

The protected Comparator result applies to the unchanged proof sources at snapshot 451872fb. Hosted Palomar verification, rendering and registry review have not run.

## Open proposals

The mixed weighted trace estimate, the complementary short-detector saving near 39/50, the remaining LONG range at 17/20, and the density-hypothesis endpoint at 3/4 remain open. They are not plotted or counted as proved full-density results.

## References

[1] PALOMAR-2026-10-11-000010 (v1), 11 October 2026. [2] PALOMAR-2026-10-09-000005 (v1), 9 October 2026. Click either reference number to open the registered submission.

[3] OpenAI, The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane Re s > 7/8 (30 September 2026). Official scope: lean/docs/003.md.

[Collet26] Hailey Collet, A zero-free half-plane with boundary 874861/1000000. https://haileystorm.github.io/research/cycle61-qrh-boundary/

[Liu26] Baiying Liu, Improved zero-free half-planes for the quasi-Riemann hypothesis, arXiv:2610.12234v2 (12 October 2026). https://arxiv.org/abs/2610.12234v2

Tao, Trudgian and Yang, arXiv:2501.16779, Table 2; Kerr, arXiv:1909.12075, Theorem 6, as identified in the checked source inventory. Exact source references and theorem identifiers are retained in BOUNDS.json and README_DRAFT.md.